IamCSC: Intuitive Assimilation Modality Driven Crossmodal Subspace Clustering for Land-Cover Identification and Hyperspectral-LiDAR Fusion

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***Abstract*— Hyperspectral (HS) and light detection and ranging (LiDAR) fusion for land-cover identification in multimodal tasks is always restricted owing to modality heterogeneity, especially when annotated samples are unavailable. Clustering techniques become pivotal in these unsupervised scenarios, yet their inability to offer interpretable fusion mechanisms poses challenges in pursuing enhanced identification performance. In this article, an Intuitive Assimilation Modality-driven Crossmodal Subspace Clustering (IamCSC) method is proposed to exploit intuitive crossmodal information. Different from existing multiview clustering approaches, our core innovation is that IamCSC distills an intuitive assimilation modality (IAM) with distinguishing geological characteristics of HS and LiDAR images simultaneously. The IAM with a small number of IAM channels still preserves the spectral similarity and diversity of neighboring samples. Meanwhile, it reflects sample elevations by peak regions along the IAM channel dimension. Specifically, the IAM connects and balances these two RS modalities to learn an assimilation cluster structure from modality-specific subspace representations. Several matrix-wise constraints are considered for capturing the optimal modality-shared representation. Given that IamCSC does not exhibit over-reliance on spatial neighborhood contributions, it achieves satisfactory performance particularly in image pairs with larger ground sampling distances (GSDs). Multiple experiments clearly prove the effectiveness of IamCSC over state-of-the-art techniques qualitatively and quantitatively. The codes are provided in https://github.com/GEOywb/IamCSC.**

***Index Terms*— Assimilation modality, hyperspectral (HS), land-cover identification, light detection and ranging (LiDAR), subspace clustering (SC).**

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I. Introduction

REMOTE sensing (RS) techniques acquire object information via remote sensors on satellites from a long distance [1], [2]. Reflected or emitted energy is captured and recorded by RS equipment to describe Earth systems in real-world tasks. The rapid evolution of RS imaging theories enables the capture of diverse useful RS modalities, including hyperspectral (HS) and light detection and ranging (LiDAR) images, for reflecting detailed object characteristics from discrepant perspectives [3]. Specifically, HS images consist of hundreds of HS channels, recording object spatial arrangements and spectral curves simultaneously [4], [5]. Their spectral ranges always extend beyond the visible spectrum [6], covering the ultraviolet and infrared spectra [7]. In contrast, LiDAR images record detailed target elevations along the vertical dimension and are not tied to spectral features [8], [9]. For accurately detecting on-ground objects, digital surface models (DSMs) can be derived from LiDAR images to describe terrain and artificial surfaces. In terms of their specific properties, fusing both RS modalities has enormous potential in accurate land-cover identification.

Existing land-cover identification techniques can be categorized into supervised, semi-supervised, and unsupervised methods in RS-related fields in terms of their learning patterns [10], [11], [12]. The first two types are generally based on classic models formerly, e.g., support vector machine and random forest, and deep neural networks (DNNs) recently [13], [14], [15], [16], [17]. Peng et al. [18] initially proposed one novel multiview hierarchical network (MVHN) technique for land-cover identification. Zhang et al. [19] enhanced crossmodal feature extraction by utilizing a cascaded block design along with a spectral-spatial attention mechanism. On the whole, the aforementioned methods rely on annotated samples for optimizing their vital classifiers. Nonetheless, in real-world tasks, sample annotation is definitely a laborious and time-consuming assignment since both HS and LiDAR images exceed the limitations of human visual perception capabilities [20]. Thus, annotated samples are always scarce or even unavailable for model training, resulting in poor supervised and semi-supervised performance [21]. In contrast, unsupervised identification, namely clustering, is capable of dividing samples into several clusters automatically without using any annotated samples [22]. This type plays a crucial

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role in complicated RS capturing scenarios, e.g., crop condition monitoring and pesticide detection [23].

Classic clustering methods are generally based on centroid, density, hierarchy, and spectral [24], [25]. Compared to the first three types (i.e., the methods based on centroid, density, and hierarchy), the last one exhibits greater distribution adaptability and requires less computation, enhancing its benefits and expanding its range of application scenarios [26], [27]. Notably, spectral-based approaches contain two stages, i.e., coefficient matrix construction and spectral clustering. Extensive studies, e.g., subspace clustering (SC) [28], generate great coefficient matrices. SC represents each sample as one sparse combination of other samples, trying to find a lower dimensional representation in underlying subspaces. Despite its great potential in computer vision, some research shows that directly using SC in RS-related fields tends to be interfered with for several reasons, e.g., inevitable noise, complex subspace structure, and sample diversity, leading to unsatisfactory clustering performance.

To confront these obstacles, multiple prior-based variants are designed [29]. Diverse variants analyze features from single modalities [30], [31]. Huang et al. [32] balanced conventional and deep SC methods by constructing a self-expressive learning network with graph constraints. Chen et al. [33] used a diffusion SC model to extract intrinsic distribution information. Considering strong spectral-spatial correlations are always neglected, Huang et al. [34] presented a Tucker decomposition-based SC model. These authors further designed a structural SC method, utilizing the low-rank HS property [35]. Furthermore, to fuse multiple modalities, some multiview SC variants perform well when exchanging modality information for accurate clustering results [36]. Zhang et al. [37] provided a low-rank tensor regularized views recovery (LATER) method to capture missing view information. Chen et al. [38] proposed a multiview SC with grouping effect (MvSCGE) to highlight the space-wise grouping effect. Considering most multiview variants ignore view consistent information, [39], [40] learned shared latent subspaces by low-rank and iterative strategies, respectively. Furthermore, [41] constructed a hypergraph regularized multiview SC model for aligning HS multiviews. Despite its satisfactory performance, this technique does not work using two absolutely different RS modalities, e.g., HS and LiDAR images, limiting its potential in real-world tasks.

Apparently, in multimodal land-cover identification tasks, these clustering methods primarily face two challenges, including ***Modality Heterogeneity*** and ***Clustering Explainability***.

1) *Modality Heterogeneity:* In this work, our motivation is to construct a crossmodal SC model for land-cover identification and HS-LiDAR fusion without any annotated samples. Existing multiview SC approaches cannot yield satisfactory performance since these two RS modalities are severely heterogeneous. Selecting a simple and crude strategy for obtaining their common coefficient matrix tends to impede the final accuracy. Thus, exploring a novel and effective breakthrough point to solve the modality heterogeneity becomes the key point.
2) *Clustering Explainability:* Existing multiview SC approaches treat HS and LiDAR images as one ordinary modality pair, without considering their specific geological characteristics and descriptions. Moreover, some black-box DNNs further weaken their modality peculiarity, making the whole clustering process unexplainable. In this article, an assimilation modality, containing distinguishing geological characteristics of HS and LiDAR images, is distilled to enhance the clustering explainability and facilitate RS modality assimilation.

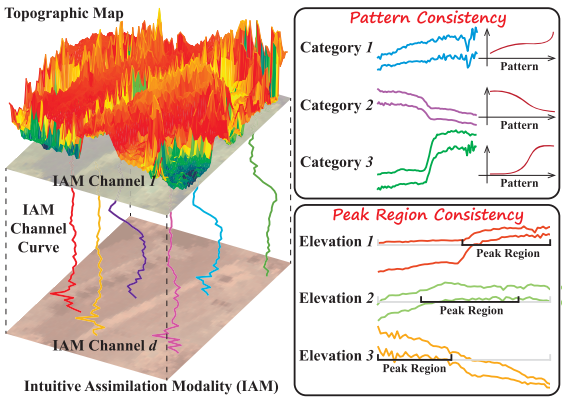


Fig. 1. Core innovation of IamCSC. The presented constraints on the IAM channel curves contribute to reflecting HS and LiDAR geological characteristics in the distilled IAM concerning pattern and peak region consistencies.

To tackle these challenges, an Intuitive Assimilation Modality-driven Crossmodal SC (IamCSC) method is proposed to exploit intuitive crossmodal information using unannotated HS and LiDAR samples. The core innovation is that, compared to existing multiview SC techniques, IamCSC distills an intuitive assimilation modality (IAM) with distinguishing geological characteristics of both HS and LiDAR images simultaneously. This presented clustering concept effectively removes the effect of modality heterogeneity and enhances clustering explainability. Specifically, as shown in Fig. 1, the distilled IAM intuitively reflects the geological characteristics of both RS modalities: 1) although the IAM channel number is smaller than the HS channel number, the distilled IAM still preserves spectral similarity and diversity of neighboring samples and 2) the distilled IAM reflects sample elevations by peak regions along the IAM channel dimension, restricting the on-ground object description ability according to their elevations. Notably, the used elevations are provided by the LiDAR-derived DSM, meaning that sample annotations are not needed. Thus, IamCSC is absolutely unsupervised and contributes to distilling the IAM by unannotated HS and LiDAR samples. IamCSC constructs one compound gradient-based HS simulation (CG-HS) constraint in terms of pattern consistency and one expanding elevation-driven LiDAR simulation (EE-LiDAR) constraint in terms of peak region consistency, which are connected and balanced by the IAM for an assimilation cluster structure. Both constraints restrict the distilled IAM and contribute to eliminating the modality heterogeneity by guiding the distillation procedure. Moreover, several nuclear and l_1 norms-based matrix-wise constraints contribute to capturing the optimal modality-shared representation. Given that IamCSC does not excessively rely on spatial neighborhood contributions, it achieves satisfactory performance, particularly in image pairs with larger ground sampling distances (GSDs). The main contributions are summarized as follows.

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1) The proposed SC method called IamCSC for land-cover identification and HS-LiDAR fusion without annotated samples is capable of distilling one IAM with distinguishing HS and LiDAR characteristics from the geologic perspective. The distilled IAM is regarded as the shared information source and is capable of being mapped back to the original RS modalities by coefficient matrices. To the best of our knowledge, it is the first work that distills one IAM for enhancing the RS modality consistency and clustering explainability in SC frameworks.
2) The IAM distilled by IamCSC preserves spectral similarity and diversity although the IAM channel number is smaller than the HS channel number. Simultaneously, the IAM restricts the on-ground description ability according to the elevations and reflects these elevations by peak regions along the IAM channel dimension. These intuitive geological characteristics are preserved and emphasized by the CG-HS and EE-LiDAR constraints in IamCSC.
3) IamCSC highlights the common coefficient sparsity of both modality-specific coefficient matrices by nuclear norm and weakens residual matrices of both RS modalities by l_1 norm. All matrix-wise constraints contribute to capturing the optimal modality-shared representation by deriving the augmented Lagrangian function in the alternating direction method of multipliers (ADMMs).

Section II provides preliminaries and related works. Section III describes IamCSC in detail. Section IV records the detailed experimental results and analysis. Section V is the conclusion.

II. Preliminaries and Related Works

A. Single-view SC Model for HS Images

The HS image is denoted as $\mathbf{X}_1 \in \mathbb{R}^{b\times m\times n}$, where m, n, and b are the length, width and HS channel number of $\mathbf{X}_1$. Single-view SC model divides all samples into clusters by defining the self-representation sparse coding function as

$$\arg\min \Psi\left(\hat{\mathbf{X}}_1 - \hat{\mathbf{X}}_1\mathbf{W}_1\right) + \beta\Theta(\mathbf{W}_1), \quad \text{s.t. } \operatorname{diag}(\mathbf{W}_1) = 0 \quad (1)$$

where $\mathbf{W}_1 \in \mathbb{R}^{mn\times mn}$ is the coefficient matrix, $\hat{\mathbf{X}}_1 \in \mathbb{R}^{b\times mn}$ is the flattened HS image, mn is the sample number, $\Psi(\cdot)$ is the given loss function, e.g., $||\cdot||_1$ and $||\cdot||_F^2$, $\Theta(\mathbf{W}_1)$ is the prior-based constraint on $\mathbf{W}_1$, e.g., $||\mathbf{W}_1||_1$ and $||\mathbf{W}_1||_*$, and β is a trade-off parameter, balancing $\Psi(\cdot)$ and $\Theta(\mathbf{W}_1)$. Notably, $\operatorname{diag}(\mathbf{W}_1) = 0$ represents $\mathbf{W}_1(i,i) = 0$ $(1 \le i \le mn)$. For one matrix $\mathbf{A}$, its l_1 norm $||\mathbf{A}||_1 = \sum_i \sum_j |\mathbf{A}(i,j)|$, its Frobenius norm $||\mathbf{A}||_F^2 = \sum_i \sum_j \mathbf{A}^2(i,j)$ and its kernel norm $||\mathbf{A}||_*$ is the sum of its singular values.

Subsequently, $\mathbf{W}_1$ generates a symmetric coefficient matrix as $\mathbf{W} = (|\mathbf{W}_1| + |\mathbf{W}_1^T|)/2$, where $\mathbf{W}_1^T$ is the transpose of $\mathbf{W}_1$. Afterward, $\mathbf{W}$ is utilized for spectral clustering as: 1) obtaining the c smallest eigenvalues of the Laplacian matrix $\mathbf{L}$ generated by $\mathbf{W}$; 2) calculating the corresponding c eigenvectors as $[\mathbf{v}_1, \ldots, \mathbf{v}_c]$; and 3) applying k-means clustering to the combined eigenvector matrix, where c is the cluster number.

Inspired by this derivation, several advanced single-view SC versions with effective constraints have been presented for HS images. Das et al. [42] introduced one unique $l_{p,q}$ norm instead of classic sparsity measures to identify inherent nonlinearity. Zhang et al. [43] extended SC in the non-Euclidean domain and considered hyperedges for better representations. Moreover, [44], [45] constructed autoencoder-based models based on self-supervision and superpixel pooling, respectively. Nonetheless, in multimodal land-cover identification scenarios, these methods constrain the information source diversity.

B. Multiview SC Model in RS Related Fields

The multiview images are denoted as $\{\mathbf{X}_i \in \mathbb{R}^{b_i\times m\times n}\}_{i=1}^K$, where K is the image source number. These images can be flattened as $\{\hat{\mathbf{X}}_i \in \mathbb{R}^{b_i\times mn}\}_{i=1}^K$. For each RS image $\hat{\mathbf{X}}_i$, its samples can be represented by the linear combination of other samples as $\hat{\mathbf{X}}_i = \hat{\mathbf{X}}_i\mathbf{W}_i + \epsilon$, where ϵ is the fitting error. Thus, the self-representation sparse coding function is formulated as

$$\arg\min \sum_{i=1}^{K} \Psi\left(\hat{\mathbf{X}}_i - \hat{\mathbf{X}}_i\mathbf{W}_i\right) + \beta\Theta(\mathbf{W}_i) \quad \text{s.t. } \operatorname{diag}(\mathbf{W}_i) = 0. \quad (2)$$

The optimal coefficient matrix set $\{\mathbf{W}_1, \ldots, \mathbf{W}_K\}$ is obtained by optimizing this function. Subsequently, the common coefficient matrix $\mathbf{W}$ for spectral clustering is generated by several restrictions. It illustrates that multiview SC models capture discriminant information from several data sources simultaneously, resulting in superior clustering performance than single-view ones. Ma et al. [46] considered linear discriminant analysis and nonnegative matrix factorization in one multiview unified framework. In RS-related fields, although multiview images can be captured by multiple sensors, their modality heterogeneity always hinders the clustering performance improvement. Thus, Some works tend to treat different modes in single RS modalities, e.g., HS images, as multiview images. Huang et al. [41] learned a common intrinsic cluster structure from the spatial and spectral HS modes. Lu et al. [47] proposed a collaborative representation-based dynamic ensemble selection strategy. Chen et al. [48] constructed small-size dictionaries for view-specific representation matrices.

Nonetheless, these multiview SC models achieve superior performance in homogeneous RS modality pairs rather than heterogeneous RS ones, e.g., the HS and LiDAR image pair. The reason is that these two RS modalities differ sharply in terms of both sample distributions and geological characteristics. Developing effective SC models for HS and LiDAR images becomes urgent in existing multimodal RS scenarios.

III. Proposed Method

A. Motivation

Given that the capabilities of multiview SC models are limited for HS and LiDAR images when their samples are unannotated, the motivation of this work is to construct an effective crossmodal SC model for land-cover identification and HS-LiDAR fusion specifically considering their modality-specific characteristics. The scheme of this work is shown in Fig. 2. The core innovation is that an IAM is distilled from both heterogeneous RS modalities as the common dictionary for information assimilation. To enhance the fusion explainability and clustering performance, the distilled IAM intuitively reflects the geological characteristics of HS and

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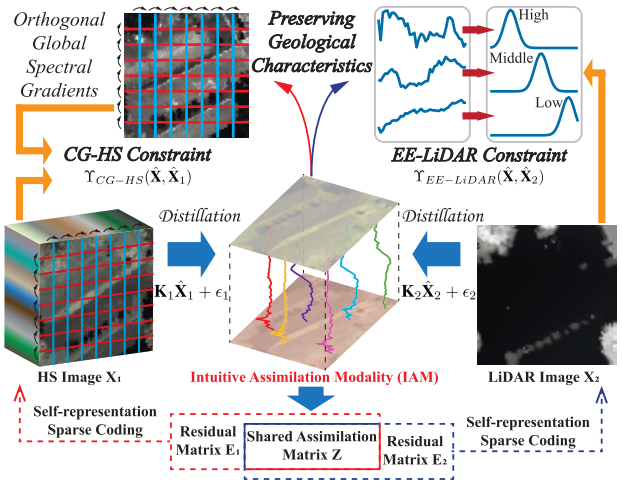


Fig. 2. Scheme of IamCSC using unannotated samples. The IAM intuitively reflects the geological characteristics of both RS modalities by constructing the CG-HS constraint in terms of pattern consistency and the EE-LiDAR constraint in terms of peak region consistency.

LiDAR images. For the HS image, its samples can be distinguished by their unique spectral curves. Thus, although the IAM channel number is smaller than the HS channel number, the IAM is supposed to preserve spectral similarity and diversity. This preservation makes the samples within the same category share similar IAM channel curves. It also promotes the IAM channel curve diversity among the samples within different categories. Notably, some RS capturing characteristics, e.g., specific absorption peaks, are not preserved in the IAM since they are unnecessary for accurate identification. For the LiDAR image, the elevations are recorded for simulating the vertical object arrangement. Accordingly, the IAM reflects these elevations by peak regions along the IAM channel dimension. To sum up, IamCSC comprehensively fuses both heterogeneous RS modalities intuitively and distills the IAM in terms of the unique geological characteristics by constraining the pattern and peak region consistencies of the IAM, respectively. These constraints together with several matrix-wise constraints improve the capability of cluster simulation.

Furthermore, the real-world RS observation mechanisms are always hampered by high orbit altitudes and limited focal lengths, leading to low resolutions and blurred details. To broaden the application, IamCSC strives to achieve satisfactory performance, particularly in RS image pairs with larger GSDs.

B. Crossmodal SC Function in IamCSC

The HS image and LiDAR derived DSM are denoted as $\mathbf{X}_1 \in \mathbb{R}^{b\times m\times n}$ and $\mathbf{X}_2 \in \mathbb{R}^{1\times m\times n}$, respectively. The DSM visualizes terrain and artificial surfaces from the elevation perspective. Although the LiDAR-derived DSM is one 2-D image, the HS image and LiDAR-derived DSM are captured by diverse RS instruments and describe land-cover objects from different perspectives. Thus, they are two different RS modalities [10]. For capturing discriminant information by networks, $\mathbf{X}_1$ and $\mathbf{X}_2$ are flattened as $\hat{\mathbf{X}}_1 \in \mathbb{R}^{b\times mn}$ and $\hat{\mathbf{X}}_2 \in \mathbb{R}^{1\times mn}$. The distilled IAM can be denoted as $\mathbf{X} \in \mathbb{R}^{d\times m\times n}$, where the IAM channel number d is smaller than the HS channel number b. Its flattened version is $\hat{\mathbf{X}} \in \mathbb{R}^{d\times mn}$. IamCSC assumes that both modalities can be transformed to $\hat{\mathbf{X}}$ by the transition matrices $\mathbf{K}_1 \in \mathbb{R}^{d\times b}$ and $\mathbf{K}_2 \in \mathbb{R}^{d\times 1}$ as

$$\hat{\mathbf{X}} = \mathbf{K}_1\hat{\mathbf{X}}_1 + \epsilon_1 = \mathbf{K}_2\hat{\mathbf{X}}_2 + \epsilon_2 \tag{3}$$

where ϵ_1 and ϵ_2 are the fitting errors. Considering the IAM $\hat{\mathbf{X}}$ represents that both heterogeneous RS modalities can be assimilated, selecting $\hat{\mathbf{X}}$ as the common crossmodal dictionary for the underlying common data structure becomes feasible and has great potential. Inspired by this assumption, the basic crossmodal SC function can be formulated as

$$\arg\min \sum_{i=1}^{2} \left|\left|\mathbf{K}_i\hat{\mathbf{X}}_i - \hat{\mathbf{X}}\mathbf{W}_i\right|\right|_F^2 + \alpha\Omega\left(\hat{\mathbf{X}}_1, \hat{\mathbf{X}}_2, \hat{\mathbf{X}}\right) + \beta\Theta(\mathbf{W}_i) \tag{4}$$

where $||\mathbf{K}_i\hat{\mathbf{X}}_i - \hat{\mathbf{X}}\mathbf{W}_i||_F^2$ contributes to minimizing both fitting errors, $\Omega(\hat{\mathbf{X}}_1, \hat{\mathbf{X}}_2, \hat{\mathbf{X}})$ represents the IAM-related constraint set for imparting distinguishing geological characteristics of HS and LiDAR images to the IAM, which is analyzed in Section III-C, $\Theta(\mathbf{W}_i)$ is the constraint on $\mathbf{W}_i$, and α and β are trade-off parameters. Notably, $\text{diag}(\mathbf{W}_i) = 0$ is preserved. In terms of the motivation, it assumes that both coefficient matrices can be decomposed into one shared low-rank assimilation matrix $\mathbf{Z}$ and two sparse residual matrices as $\mathbf{W}_i = \mathbf{Z} + \mathbf{E}_i$. Thus, $\Theta(\mathbf{W}_i)$ can be formulated as

$$\Theta(\mathbf{W}_i) = ||\mathbf{E}_i||_1 + ||\mathbf{Z}||_*, \quad \text{s.t. } \mathbf{W}_i = \mathbf{Z} + \mathbf{E}_i, \quad (i = 1, 2). \tag{5}$$

The common coefficient sparsity is highlighted by introducing one kernel norm-based constraint. By minimizing both residual matrices $\mathbf{W}_1$ and $\mathbf{W}_2$, $\mathbf{Z}$ is capable of constructing the common modality-shared subspace of HS and LiDAR images for joint analysis. After that, $\mathbf{Z}$ is used to calculate the symmetric coefficient matrix as $\mathbf{W} = (|\mathbf{Z}| + |\mathbf{Z}^T|)/2$ for spectral clustering.

C. IAM-Related Constraint Set

The basic crossmodal SC function in (4) attempts to solve the sparse and low-rank optimization problems when focusing on $\Psi(\mathbf{K}_i\hat{\mathbf{X}}_i - \hat{\mathbf{X}}\mathbf{W}_i)$ and $\Theta(\mathbf{W}_i)$. However, this procedure can be interfered with by inevitable noise. Studies prove that improved performance can be achieved when useful priors in RS modalities are taken into account.

The IAM maintains both spectral similarity and diversity of the HS image while also incorporating the LiDAR elevations simultaneously. Specifically, these distinguishing geological characteristics are treated as the fresh impetus to the IAM by considering the IAM-related constraint set $\Omega(\hat{\mathbf{X}}_1, \hat{\mathbf{X}}_2, \hat{\mathbf{X}})$, including the CG-HS and EE-LiDAR constraints.

1) CG-HS Constraint: Adjacent HS samples in the spatial domain can be identified considering their spectral similarity and diversity. These samples belong to the same category when their spectral curves are sufficiently similar. Thus, the IAM is supposed to contain this characteristic due to two reasons: 1) IamCSC attempts to make the IAM reflect the HS geological characteristic intuitively, enhancing its clustering explainability and 2) the HS samples are likely to be linearly combined by fewer IAM atoms, enhancing the coefficient sparsity. In the common crossmodal dictionary $\hat{\mathbf{X}} \in \mathbb{R}^{d\times mn}$, each column represents one atom for linear combination.

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To describe the HS sample connection within local regions, the 3-D matrix versions of $\hat{\mathbf{X}}_1$, $\hat{\mathbf{X}}_2$, and $\hat{\mathbf{X}}$, i.e., $\mathbf{X}_1$, $\mathbf{X}_2$, and $\mathbf{X}$, are considered. The anisotropic HS variation G is simulated in terms of two orthogonal global spectral gradients as

$$\begin{aligned} G^x\left[\mathbf{X}(k,i,j)\right] &= \mathbf{X}(k,i+1,j) - \mathbf{X}(k,i,j) \\ G^y\left[\mathbf{X}(k,i,j)\right] &= \mathbf{X}(k,i,j+1) - \mathbf{X}(k,i,j) \\ & 1 \le k \le d, \quad 1 \le i \le m, \quad 1 \le j \le n \end{aligned} \tag{6}$$

where k, i, and j are used to locate the detailed matrix entry positions, and d is the IAM channel number. The CG-HS constraint along two orthogonal directions is

$$\begin{aligned} &\Upsilon_{\text{CG-HS}}\left(\hat{\mathbf{X}}, \hat{\mathbf{X}}_1\right) \\ &= \left| \frac{1}{d}\sum_{k=1}^{d} G^x\left[\mathbf{X}(k,i,j)\right] - \frac{1}{b}\sum_{k=1}^{b} G^x\left[\mathbf{X}_1(k,i,j)\right] \right| \\ &\quad + \left| \frac{1}{d}\sum_{k=1}^{d} G^y\left[\mathbf{X}(k,i,j)\right] - \frac{1}{b}\sum_{k=1}^{b} G^y\left[\mathbf{X}_1(k,i,j)\right] \right| \end{aligned} \tag{7}$$

where b is the $\mathbf{X}_1$ channel number. The periodic assignment strategy is used when meeting boundary rows and columns. For a more compact simulation, the 3-D matrix-wise constraints are written by considering the finite difference operators $\mathbf{H}_x \in \mathbb{R}^{n\times n}$ and $\mathbf{H}_y \in \mathbb{R}^{m\times m}$ as

$$\begin{aligned} \Upsilon_{\text{CG-HS}}\left(\hat{\mathbf{X}}, \hat{\mathbf{X}}_1\right) &= ||\mathbf{B}_d\mathbf{X}\mathbf{H}_x - \mathbf{B}_b\mathbf{X}_1\mathbf{H}_x||_F^2 \\ &\quad + \left|\left|\mathbf{B}_d(\mathbf{X})_{2\rightleftharpoons 3}\mathbf{H}_y - \mathbf{B}_b(\mathbf{X}_1)_{2\rightleftharpoons 3}\mathbf{H}_y\right|\right|_F^2 \end{aligned} \tag{8}$$

where $(\mathbf{X})_{i\rightleftharpoons j}$ exchanges the ith and jth dimensions of $\mathbf{X}$, the global average operator $\mathbf{B}_i$ is $[1/i, 1/i, \ldots, 1/i]_{1\times i}$. Notably, by introducing two forward finite difference operators, the CG-HS constraint in (7) evolutes from a scabrous version into an unambiguous one for iterative optimization.

Apparently, (8) shows that IamCSC captures spectral information through orthogonal gradient measurement, rather than relying on neighboring sample combination or convolution operations. Thus, IamCSC does not exhibit over-reliance on spatial neighborhood contributions. It achieves satisfactory performance in image pairs with larger GSDs although their neighborhood contributions are discrete and unreliable.

2) EE-LiDAR Constraint: The recorded LiDAR elevations contribute to simulating the vertical arrangements of objects conveniently. Concerning our motivation, the distilled IAM should reflect these LiDAR elevations by introducing exclusive restrictions on describing peak regions. Similarly, this setting is conducive to further enhancing the clustering explainability and the coefficient sparsity in linear combination when all IAM atoms contain the LiDAR geological characteristic. For one IAM sample $\hat{\mathbf{x}} \in \mathbb{R}^{d\times 1}$, its peak region can be accurately captured by one global smoothing operator $\mathbf{P} \in \mathbb{R}^{d\times d}$. Notably, $\mathbf{P}$ is one circulant matrix on the list as

$$\left\{ \overbrace{\frac{1}{\tau}\ \frac{1}{\tau}\ \cdots\ \frac{1}{\tau}}^{\tau}, \overbrace{0\ \ 0\ \ \cdots\ \ 0}^{d-\tau} \right\} \tag{9}$$

where τ is the global smoothing index and $\tau < d$. Notably, in $\mathbf{P}\hat{\mathbf{X}}$, the global peak region is accurately captured by removing the noise effect. After that, an intuitive LiDAR elevation visualization of $\hat{\mathbf{X}}$ is generated for restricting the IAM. The LiDAR image $\hat{\mathbf{X}}_2$ is initially normalized into $[0, d]$. The normalized elevation value of one LiDAR sample $\hat{x}_2$ can be regarded as the mean μ for generating its elevation normal distribution curve. For $\hat{\mathbf{X}}_2$, its intuitive LiDAR elevation visualization is formulated as

$$\Lambda\left(\hat{\mathbf{X}}_2, \sigma^2\right) = N_{\text{pdf}}\left[\sqcap_0^d\left(\hat{\mathbf{X}}_2\right), \sigma^2\right] \tag{10}$$

where $\sqcap_0^d(\cdot)$ is the $[0, d]$ normalization operator, $N_{\text{pdf}}[\cdot]$ is the normal probability density function for generating elevation normal distribution curves, and σ^2 is the curve variance. IamCSC attempts to guarantee the LiDAR geological characteristic by closing the elevation description gap between $\hat{\mathbf{X}}$ and $\hat{\mathbf{X}}_2$ as

$$\Upsilon_{\text{EE-LiDAR}}\left(\hat{\mathbf{X}}, \hat{\mathbf{X}}_2\right) = \left|\left|\mathbf{P}\hat{\mathbf{X}} - \Lambda\left(\hat{\mathbf{X}}_2, \sigma^2\right)\right|\right|_F^2. \tag{11}$$

Apparently, this constraint makes the distilled IAM reflect sample elevations along the IAM channel dimension. In the IAM, the elevations are represented in the form of peak regions rather than data values. By describing the elevations visually in (11), these elevations are intuitive profiles in the IAM rather than stiff numbers.

Algorithm 1 Iterative Variable Updating in IamCSC

Input: $\mathbf{X}_1$, $\mathbf{X}_2$, α_1, α_2, β, ε, I_{num}, and η.

Initializing $\hat{\mathbf{X}}$, $\mathbf{Z}$, $\{\mathbf{E}_i\}_{i=1}^2$, $\{\mathbf{W}_i\}_{i=1}^2$, $\{\mathbf{K}_i\}_{i=1}^2$, and $\{\mathbf{Y}_i\}_{i=1}^2$ randomly, $e_\infty = 0$, $\sigma = 1$, $\lambda = 1$, and $t = 1$;

if $t \le \varepsilon$ or $e_\infty \ge \varepsilon$ **then**

- Updating $\hat{\mathbf{X}}$ as $\hat{\mathbf{X}}^{t+1} = \hat{\mathbf{X}}^t - \eta\nabla\mathcal{L}_{\hat{\mathbf{X}}^t}$;
- Updating $\mathbf{Z}$ as $\mathbf{Z}^{t+1} = \mathbf{Z}^t - \eta\nabla\mathcal{L}_{\mathbf{Z}^t}$;
- Updating $\{\mathbf{E}_i\}_{i=1}^2$ as $\{\mathbf{E}_i^{t+1}\}_{i=1}^2 = \{\mathbf{E}_i^t\}_{i=1}^2 - \eta\nabla\mathcal{L}_{\{\mathbf{E}_i^t\}_{i=1}^2}$;
- Updating other variables using their gradients;
- Calculating e_∞, $t = t + 1$;

Output: $\mathbf{Z}$ for spectral clustering.

D. Optimization Algorithm

Inspired by the aforementioned constraints and analysis, the crossmodal SC function in IamCSC can be derived as

$$\begin{aligned} &\arg\min \sum_{i=1}^{2}\left|\left|\mathbf{K}_i\hat{\mathbf{X}}_i - \hat{\mathbf{X}}\mathbf{W}_i\right|\right|_F^2 \\ &\quad + \alpha_1\Upsilon_{\text{CG-HS}}\left(\hat{\mathbf{X}}, \hat{\mathbf{X}}_1\right) + \alpha_2\Upsilon_{\text{EE-LiDAR}}\left(\hat{\mathbf{X}}, \hat{\mathbf{X}}_2\right) \\ &\quad + \beta(||\mathbf{E}_i||_1 + ||\mathbf{Z}||_*), \quad \text{s.t. } \mathbf{W}_i = \mathbf{Z} + \mathbf{E}_i \quad (i = 1, 2) \\ &= \arg\min \sum_{i=1}^{2}\left|\left|\mathbf{K}_i\hat{\mathbf{X}}_i - \hat{\mathbf{X}}\mathbf{W}_i\right|\right|_F^2 \\ &\quad + \alpha_1||\mathbf{B}_d\mathbf{X}\mathbf{H}_x - \mathbf{B}_b\mathbf{X}_1\mathbf{H}_x||_F^2 \\ &\quad + \alpha_1\left|\left|\mathbf{B}_d(\mathbf{X})_{2\rightleftharpoons 3}\mathbf{H}_y - \mathbf{B}_b(\mathbf{X}_1)_{2\rightleftharpoons 3}\mathbf{H}_y\right|\right|_F^2 \\ &\quad + \alpha_2\left|\left|\mathbf{P}\hat{\mathbf{X}} - \Lambda\left(\hat{\mathbf{X}}_2, \sigma^2\right)\right|\right|_F^2 \\ &\quad + \beta(||\mathbf{E}_i||_1 + ||\mathbf{Z}||_*), \quad \text{s.t. } \mathbf{W}_i = \mathbf{Z} + \mathbf{E}_i \quad (i = 1, 2) \end{aligned} \tag{12}$$

where α_1 and α_2 are two trade-off parameters. Apparently, this function does not have analytical solutions and is regarded as an intractable problem to be optimized. In other words, the whole optimization problem can be split into multiple

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subproblems and solved by the ADMMs. Inspired by this optimization strategy, the augmented Lagrangian function of (12) is obtained as

$$\begin{aligned}\mathcal{L} = &\sum_{i=1}^{2}\left|\left|\mathbf{K}_i\hat{\mathbf{X}}_i - \hat{\mathbf{X}}\mathbf{W}_i\right|\right|_F^2 \\ &+ \alpha_1||\mathbf{B}_d\mathbf{X}\mathbf{H}_x - \mathbf{B}_b\mathbf{X}_1\mathbf{H}_x||_F^2 \\ &+ \alpha_1\left|\left|\mathbf{B}_d(\mathbf{X})_{2=3}\mathbf{H}_y - \mathbf{B}_b(\mathbf{X}_1)_{2=3}\mathbf{H}_y\right|\right|_F^2 \\ &+ \alpha_2\left|\left|\mathbf{P}\hat{\mathbf{X}} - \Lambda\left(\hat{\mathbf{X}}_2, \sigma^2\right)\right|\right|_F^2 \\ &+ \beta(||\mathbf{E}_i||_1 + ||\mathbf{Z}||_*) + \lambda/2||\mathbf{W}_i - \mathbf{M}_i||_F^2 \end{aligned} \tag{13}$$

where λ is one trade-off parameter, $\{\mathbf{M}_i\}_{i=1}^2 = \mathbf{Z}+\mathbf{E}_i-\mathbf{Y}_i/\lambda$, $\{\mathbf{Y}_i\}_{i=1}^2$ are Lagrangian multipliers. All variables in (13) are iteratively updated by fixing the other variables. The subproblem for $\hat{\mathbf{X}}$ is formulated as

$$\begin{aligned}\arg\min &\sum_{i=1}^{2}\left|\left|\mathbf{K}_i\hat{\mathbf{X}}_i - \hat{\mathbf{X}}\mathbf{W}_i\right|\right|_F^2 \\ &+ \alpha_1||\mathbf{B}_d\mathbf{X}\mathbf{H}_x - \mathbf{B}_b\mathbf{X}_1\mathbf{H}_x||_F^2 \\ &+ \alpha_1\left|\left|\mathbf{B}_d(\mathbf{X})_{2=3}\mathbf{H}_y - \mathbf{B}_b(\mathbf{X}_1)_{2=3}\mathbf{H}_y\right|\right|_F^2 \\ &+ \alpha_2\left|\left|\mathbf{P}\hat{\mathbf{X}} - \Lambda\left(\hat{\mathbf{X}}_2, \sigma^2\right)\right|\right|_F^2. \end{aligned} \tag{14}$$

Equation (14) can be simplified as $\arg\min(\mathcal{L}^1+\alpha_1\mathcal{L}^2+\alpha_2\mathcal{L}^3)$. Its gradient calculation satisfies the superposition principle. The gradient of $\mathcal{L}^1(\hat{\mathbf{X}})$ with respect to $\hat{\mathbf{X}}$ is obtained as

$$\nabla\mathcal{L}_{\hat{\mathbf{X}}}^1 = \sum_{i=1}^{2}\hat{\mathbf{X}}\mathbf{W}_i\mathbf{W}_i^T - \mathbf{K}_i\hat{\mathbf{X}}_i\mathbf{W}_i^T. \tag{15}$$

The gradient of $\mathcal{L}^2$ with respect to $\hat{\mathbf{X}}$ is obtained as

$$\begin{aligned}\nabla\mathcal{L}_{\hat{\mathbf{X}}}^2 = &\mathbf{B}_d\mathbf{B}_d^T\mathbf{X}[:]\mathbf{H}_x\mathbf{H}_x^T - \mathbf{B}_d\mathbf{B}_b^T\mathbf{X}_1[:]\mathbf{H}_x\mathbf{H}_x^T \\ &+ \mathbf{B}_d\mathbf{B}_d^T(\mathbf{X}[:])_{2=3}\mathbf{H}_y\mathbf{H}_y^T - \mathbf{B}_d\mathbf{B}_b^T(\mathbf{X}_1[:])_{2=3}\mathbf{H}_y\mathbf{H}_y^T \end{aligned} \tag{16}$$

where $\mathbf{X}[:]$ reminds that $\hat{\mathbf{X}}$ is reshaped into $\mathbf{X} \in \mathbb{R}^{d\times m\times n}$. For $\mathcal{L}^3$, its gradient with respect to $\hat{\mathbf{X}}$ is obtained as

$$\nabla\mathcal{L}_{\hat{\mathbf{X}}}^3 = \mathbf{P}^T\mathbf{P}\hat{\mathbf{X}} - \mathbf{P}\Lambda\left(\hat{\mathbf{X}}_2, \sigma^2\right). \tag{17}$$

Thus, the total gradient of $\mathcal{L}$ with respect to $\hat{\mathbf{X}}$ is obtained as

$$\nabla\mathcal{L}_{\hat{\mathbf{X}}} = \nabla\mathcal{L}_{\hat{\mathbf{X}}}^1 + \alpha_1\nabla\mathcal{L}_{\hat{\mathbf{X}}}^2 + \alpha_2\nabla\mathcal{L}_{\hat{\mathbf{X}}}^3. \tag{18}$$

The objective function for $\mathbf{Z}$ is obtained as

$$\arg\min\beta||\mathbf{Z}||_* + \sum_{i=1}^{2}\lambda/2||\mathbf{W}_i - \mathbf{M}_i||_F^2. \tag{19}$$

To solve this sub-problem, $\mathbf{Z}$ is decomposed by singular value decomposition as $\mathbf{Z} = \mathbf{U}\mathbf{S}\mathbf{V}^T$, where $\mathbf{U}$ and $\mathbf{V}$ are singular matrices, $\mathbf{S}$ is one diagonal matrix with all singular values. Then, the derivative of $\mathbf{Z}$ can be denoted as $\mathbf{U}\mathbf{V}^T$, which is directly set to zero. To summarize, $\nabla\mathcal{L}_{\mathbf{Z}}$ is obtained as

$$\nabla\mathcal{L}_{\mathbf{Z}} = \beta\mathbf{U}\mathbf{V}^T + \sum_{i=1}^{2}\lambda/2(\mathbf{M}_i - \mathbf{W}_i) \tag{20}$$

where $\mathbf{U}$ and $\mathbf{V}^T$ are decomposed by $\mathbf{Z}$. The corresponding objective function with respect to $\{\mathbf{E}_i\}_{i=1}^2$ is formulated as

$$\left\{\arg\min\beta||\mathbf{E}_i||_1 + \lambda/2||\mathbf{W}_i - \mathbf{M}_i||_F^2\right\}_{i=1}^2. \tag{21}$$

Thus, the gradient $\{\nabla\mathcal{L}_{\mathbf{E}_i}\}_{i=1}^2$ is obtained as

$$\left\{\nabla\mathcal{L}_{\mathbf{E}_i}\right\}_{i=1}^2 = \beta\mathrm{sgn}(\mathbf{E}_i) + \lambda/2(\mathbf{M}_i - \mathbf{W}_i). \tag{22}$$

Meanwhile, the gradients of $\mathcal{L}$ with respect to other variables, including $\mathbf{W}_i$, $\mathbf{K}_i$, and $\mathbf{Y}_i$, can be formulated as

$$\begin{aligned}\left\{\nabla\mathcal{L}_{\mathbf{W}_i}\right\}_{i=1}^2 &= \hat{\mathbf{X}}^T\hat{\mathbf{X}}\mathbf{W}_i - \hat{\mathbf{X}}^T\mathbf{K}_i\hat{\mathbf{X}}_i + \lambda/2(\mathbf{W}_i - \mathbf{M}_i) \\ \left\{\nabla\mathcal{L}_{\mathbf{K}_i}\right\}_{i=1}^2 &= \mathbf{K}_i\hat{\mathbf{X}}_i\hat{\mathbf{X}}_i^T - \hat{\mathbf{X}}\mathbf{W}_i\hat{\mathbf{X}}_i^T \\ \left\{\nabla\mathcal{L}_{\mathbf{Y}_i}\right\}_{i=1}^2 &= \mathbf{Y}_i/(2\lambda) + \mathbf{W}_i - \mathbf{Z} - \mathbf{E}_i/2. \end{aligned} \tag{23}$$

Algorithm 1 provides a detailed illustration of the whole iterative variable updating procedure for (12). The learning rate η controls the convergence speed. Generally, larger values of η correspond to higher efficiencies but unstable clustering performance. The updating loss $e_\infty = ||\mathbf{X}^{t+1} - \mathbf{X}^t||_\infty$ is calculated for optimization. All variables are iteratively updated until $e_\infty < \varepsilon$ or the iteration number iter becomes bigger than I_{num}. The time complexity of Algorithm 1 is fully analyzed for verifying the efficiency of IamCSC. Specifically, the time complexity for $\nabla\mathcal{L}_{\hat{\mathbf{X}}}$ and $\{\nabla\mathcal{L}_{\mathbf{K}_i}\}_{i=1}^2$ are $\mathcal{O}(dm^2n^2)$ since mn is bigger than both b and d, for $\nabla\mathcal{L}_{\mathbf{Z}}$ and $\{\nabla\mathcal{L}_{\mathbf{W}_i}\}_{i=1}^2$ are $\mathcal{O}(m^3n^3)$, for $\{\nabla\mathcal{L}_{\mathbf{E}_i}\}_{i=1}^2$ and $\{\nabla\mathcal{L}_{\mathbf{Y}_i}\}_{i=1}^2$ are $\mathcal{O}(m^2n^2)$ in terms of matrix addition and can be neglected. In conclusion, the time complexity of Algorithm 1 is primarily determined by the sample number mn and the iteration number iter.

IV. Experimental Results and Analysis

A. Dataset Description

1) Houston 2018 Dataset: Was captured above Houston University by the National Center for Airborne Laser Mapping (NCALM). It provides HS images, LiDAR point cloud data, intensity rasters from the first return per channel, and LiDAR-derived DSMs. The HS images span a wavelength range from 380 to 1050 nm with 48 HS channels. Only the LiDAR-derived DSMs were used to provide sample elevations for IamCSC. The LiDAR point cloud data and intensity rasters are not used. There are 20 land-cover categories. Due to memory constraints, two subscene pairs with 300 × 300 samples, e.g., S_1 and S_2, were chosen. To validate the effectiveness of IamCSC on low-quality RS images, systematic sampling was employed to capture S_{1H} at a relatively small GSD of 3 m, while S_{1L} and S_{2L} were captured at a relatively large GSD of 6 m. Large GSDs correspond to low resolutions and blurred details.

2) MUUFL Dataset: was captured above the gulf park campus at the University of Southern Mississippi by the reflective optics system imaging spectrometer sensor. Its HS image has 64 HS channels and was captured by the ITRES Compact Airborne Spectrographic Imager (CASI-1500) sensor. Its LiDAR image (LiDAR derived DSM) was captured by airborne laser terrain mapping (ALTM). One subscene pair M with 200 × 200 samples was chosen. Systematic sampling was used for generating M_L at a 4 m GSD. Their visualizations and sampling versions are shown in Fig. 3.

B. Experimental Setting

One PC with 16 GB RAM (Intel Core i7-8750 CPU) was used. The codes were implemented by MATLAB R2016a.

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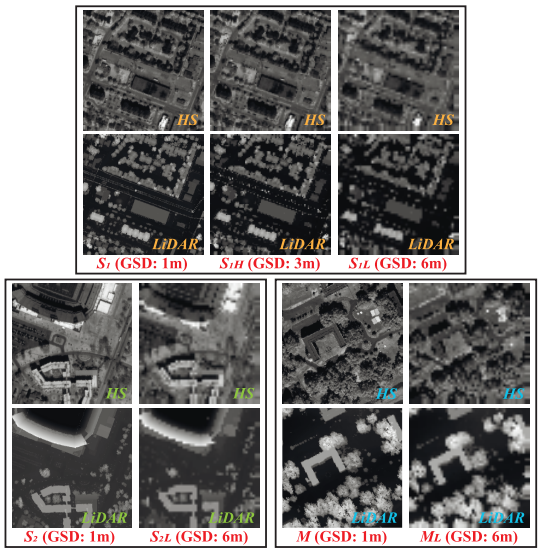


Fig. 3. Visualizations of all subscene pairs, i.e., S_1, S_2, and M, and their sampling versions, i.e., S_{1H}, S_{1L}, S_{2L}, and M_L.

Before all experiments, the HS and LiDAR images were both normalized into [0, 1]. The IAM channel number d was set as 40. Section III-B shows that d should be smaller than b in IamCSC since the IAM is expected to have fewer channels than the HS image. Its effect on clustering performance is analyzed in Section IV-D. The global smoothing index τ and the curve variance σ^2 were set as 3 and 1, respectively. The trade-off parameters α_1, α_2, β, and λ were set as 1, 1, 0.5, and 0.1, respectively. The initial learning rate η was set as 10^{-5} and was multiplied by 0.5 at each one hundred epochs. The parameter ε and the maximum iteration number I_{num} were set as 10^{-10} and 2000. The matrices to be optimized were randomly initialized. The sample annotations were not used since IamCSC is absolutely unsupervised. Five evaluation metrics, including the accuracy per category, average accuracy (AA), overall accuracy (OA), purity (PT), and normalized mutual information (NMI), were used to measure all approaches [49]. The AA is the average of all accuracies per category. The OA is regarded as the primary metric since it is the most important one for land-cover identification. These metrics are formulated as

$$\text{AA} = \frac{1}{c}\sum_{i}^{c}\frac{g_{\text{correct}}^{i}}{g_{\text{all}}^{i}}, \quad \text{OA} = \frac{g_{\text{correct}}}{g_{\text{all}}}$$

$$\text{PT} = \frac{1}{g_{\text{all}}}\sum_{j}\max_{i}\left|\succcurlyeq_j \cap \ltimes_i\right|$$

$$\text{NMI} = 2 \times \text{MI}(\mathbb{M}, \mathbb{N})/[H(\mathbb{M}) + H(\mathbb{N})] \tag{24}$$

where g_{correct} and g_{all} are the correctly clustered and total sample numbers, g_{correct}^{i} and g_{all}^{i} are the correctly clustered and total sample numbers in the ith real category, $\mathbb{M} = [\succcurlyeq_1, \succcurlyeq_2, \cdots, \succcurlyeq_c]$ and $\mathbb{N} = [\ltimes_1, \ltimes_2, \cdots, \ltimes_c]$ are the clustered and real categories, $\text{MI}(\cdot)$ and $H(\cdot)$ are the mutual information and cross entropy, and c is the real category number. For a fair and clear comparison, the clustered category number was set as c for matching [34]. The clustered and real categories were matched by the advanced Hungarian algorithm[1] [35].

[1]http://www.cad.zju.edu.cn/home/dengcai/Data/Clustering.html

Diverse state-of-the-art clustering techniques were chosen for comparison, including single-view clustering techniques: k-means, spectral clustering (SPC), sparse SC (SSC), clustering by aggregation with sorting-based indexing and explainability (CLASSIX) [27], bipartite graph-based projected clustering (BGPC) [22], dual graph autoencoder (DGAE) [26], joint unmixing and anomaly detection network (UADNet) [21], 3-D regularized SSC (3DSSC) [29], and multiview clustering techniques: robust spectral embedding completion-based incomplete multiview clustering (SEC-IMVC) [30], enhanced tensor low-rank and sparse representation recovery (ETLSRR) [31], deep complementary and consensus nonnegative matrix factorization (DCCNMF) [23], LATER [37], efficient one-pass multiview SC (EOMSC) [36], and MvSCGE [38]. Single-view clustering techniques use the HS image. The descriptions of other techniques are provided as

1) *Single-view Clustering Techniques:*
   a) *CLASSIX:* Consists of one greedy aggregation phase and one group merging procedure for images.
   b) *BGPC:* Alleviates noise interference and initializes anchors by images in each generated superpixel.
   c) *DGAE:* Embeds the geometric structure into feature extraction directly by considering sample relationships.
   d) *UADNet:* Has a novel feedback mechanism in its clustering network between the upstream and downstream processes.
   e) *3-DSSC:* Takes into account the spatial information of images by incorporating regularization terms into SSC models.
2) *Multiview Clustering Techniques:*
   a) *SEC-IMVC:* Combines SEC and discrete cluster indicator learning into one unified framework.
   b) *ETLSRR:* Treats multiview clustering as tensor representation recovery and similarity graph learning problems.
   c) *DCCNMF:* Balances nonnegative matrix factorization and coupled nonnegative matrix factorization in multiview data.
   d) *LATER:* Reconstructs missing views and captures multilevel graphs in one unified framework for comprehensive learning.
   e) *EOMSC:* Generates the clustering labels directly with optimal anchor graph for reducing memory consumption.
   f) *MvSCGE:* Preserves the locality in subspaces and highlights subspacewise grouping effects.

C. Experimental Results and Comparison

The results on S_{1H}, S_{1L}, S_{2L}, and M_L are recorded in Tables I–IV, respectively. Generally, clustering performance is not as good as classification performance due to the lack of sample annotations. The high orbit altitudes and limited focal lengths in HS and LiDAR images further add to the clustering difficulty, making it challenging to achieve great improvements in accuracy when using RS modalities. The spatial information discontinuity caused by low spatial resolution also exacerbates the challenges of clustering. Specifically, due to large GSDs, some RS samples cannot be correctly clustered and

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TABLE I

EVALUATION METRICS, INCLUDING THE ACCURACY PER CATEGORY (THE FIRST FIVE ROWS), OA, AA, PT, AND NMI, ON S_{1H} WITH A RELATIVELY SMALL GSD IN THE HOUSTON 2018 DATASET. THE BEST METRICS ARE HIGHLIGHTED IN BOLD. IAMCSC OUTPERFORMS OTHER TECHNIQUES IN TERMS OF ALL METRICS COMPREHENSIVELY

Evaluation Metrics	kmeans	SPC	SSC	CLASSIX	BGPC	DGAE	UADNet	3DSSC	SEC-IMVC	ETLSRR	DCCNMF	LATER	EOMSC	MvSCGE	IamCSC
Category 1	72.73	79.72	94.41	55.94	0.00	53.15	13.99	74.13	95.10	89.51	9.09	95.80	62.94	**96.50**	78.32
Category 2	87.00	89.51	75.90	85.65	**97.77**	67.43	95.46	58.29	75.76	66.89	35.27	62.02	88.22	43.74	82.53
Category 3	43.64	3.64	23.22	12.17	5.87	19.44	4.90	3.36	24.48	2.80	34.27	32.73	2.66	**66.43**	36.62
Category 4	0.00	0.00	59.67	1.98	5.12	6.45	6.12	65.12	46.28	**73.55**	33.06	36.86	0.00	49.75	48.60
Category 5	59.49	76.99	33.14	82.97	69.16	86.19	78.25	53.16	49.94	21.40	40.39	48.56	**95.05**	54.55	58.46
AA (%)	52.57	49.97	57.27	47.74	35.58	46.53	39.74	50.81	58.31	50.83	30.42	55.19	49.77	**62.19**	60.91
OA (%)	58.23	55.95	54.37	56.84	55.61	52.48	57.29	48.49	56.29	46.39	34.92	50.72	58.76	53.40	**62.88**
PT (÷100)	59.31	56.68	58.44	58.44	59.52	56.97	59.02	60.15	58.34	50.14	42.58	51.56	59.73	58.62	**62.88**
NMI (÷100)	34.07	32.99	29.45	34.12	34.98	25.37	32.40	29.98	30.10	25.06	9.66	16.78	**40.96**	38.33	27.89

TABLE II

EVALUATION METRICS, INCLUDING THE ACCURACY PER CATEGORY (THE FIRST FIVE ROWS), OA, AA, PT, AND NMI, ON S_{1L} WITH A RELATIVELY LARGE GSD IN THE HOUSTON 2018 DATASET. THE BEST METRICS ARE HIGHLIGHTED IN BOLD. IAMCSC ACHIEVES SATISFACTORY PERFORMANCE PARTICULARLY IN IMAGE PAIRS WITH LARGER GSDS

Evaluation Metrics	kmeans	SPC	SSC	CLASSIX	BGPC	DGAE	UADNet	3DSSC	SEC-IMVC	ETLSRR	DCCNMF	LATER	EOMSC	MvSCGE	IamCSC
Category 1	81.08	86.49	83.78	0.00	0.00	51.35	13.51	91.89	91.89	**94.59**	67.57	83.78	10.91	91.89	91.89
Category 2	89.95	88.86	60.05	81.79	**97.83**	54.35	96.74	54.08	80.16	66.03	32.88	60.33	72.28	42.93	86.96
Category 3	3.43	3.43	**45.71**	38.86	5.71	13.71	4.57	16.57	21.14	2.29	20.57	34.86	29.14	44.57	29.71
Category 4	0.67	0.00	42.28	2.68	**75.84**	34.90	7.38	58.39	42.28	17.45	12.08	30.20	0.00	53.02	9.40
Category 5	85.78	76.61	55.50	78.90	21.56	88.53	77.06	55.96	47.25	66.06	27.52	52.29	**97.25**	57.34	75.23
AA (%)	52.18	51.08	57.47	40.45	40.19	48.57	39.85	55.38	56.55	49.28	32.12	52.29	41.90	57.95	**58.64**
OA (%)	58.61	56.18	54.49	57.55	55.97	51.53	57.87	49.74	56.18	47.73	27.46	49.95	56.28	50.05	**61.67**
PT (÷100)	59.13	57.02	63.46	60.40	59.98	60.82	59.13	60.19	58.82	51.74	39.81	50.16	60.40	58.29	**63.78**
NMI (÷100)	37.24	33.79	33.75	31.57	36.21	29.99	32.11	31.04	28.89	28.21	5.26	16.42	39.38	33.06	**39.42**

TABLE III

EVALUATION METRICS, INCLUDING THE ACCURACY PER CATEGORY (THE FIRST EIGHT ROWS), OA, AA, PT, AND NMI, ON S_{2L} WITH A RELATIVELY LARGE GSD IN THE HOUSTON 2018 DATASET. THE BEST METRICS ARE HIGHLIGHTED IN BOLD. THE OBTAINED CONCLUSIONS ARE FURTHER VERIFIED

Evaluation Metrics	kmeans	SPC	SSC	CLASSIX	BGPC	DGAE	UADNet	3DSSC	SEC-IMVC	ETLSRR	DCCNMF	LATER	EOMSC	MvSCGE	IamCSC
Category 1	93.40	85.85	87.74	61.32	93.40	65.09	94.34	50.00	91.51	66.04	39.62	72.64	**96.23**	63.21	88.80
Category 2	70.37	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.48	66.67	44.44	3.70	18.52	**96.30**	92.59
Category 3	36.17	32.77	31.06	65.63	48.94	62.13	51.06	49.36	45.11	54.89	39.57	48.09	64.26	**72.34**	49.36
Category 4	0.00	53.13	43.75	46.88	0.00	**57.81**	26.56	10.94	15.63	23.44	17.19	12.50	0.00	18.75	0.00
Category 5	26.72	36.21	**41.38**	17.24	8.62	7.76	3.45	36.21	11.21	26.72	18.97	25.86	24.14	13.79	6.03
Category 6	33.61	0.84	61.34	24.37	**94.12**	67.23	24.37	64.71	68.07	26.89	20.17	52.94	60.50	59.66	88.24
Category 7	0.00	34.29	31.43	42.86	0.00	**100.00**	45.71	42.86	2.86	14.29	8.57	82.86	11.43	22.86	51.43
Category 8	42.19	54.69	65.63	33.85	**69.79**	40.10	53.65	46.88	60.42	40.10	44.79	40.63	18.23	46.35	68.75
AA (%)	37.81	37.22	45.29	36.51	39.36	50.02	37.39	37.62	47.03	39.88	29.17	42.40	36.66	49.16	**55.64**
OA (%)	39.71	40.49	50.56	42.28	52.57	50.67	43.51	44.74	49.89	42.17	32.77	44.63	44.41	51.34	**55.59**
PT (÷100)	53.58	54.03	63.42	50.67	58.72	63.31	50.34	58.50	59.17	50.11	43.06	54.03	46.53	62.19	**67.67**
NMI (÷100)	35.30	38.20	44.84	30.21	48.77	49.77	29.82	43.52	37.73	32.26	17.63	32.34	35.99	46.96	**50.09**

TABLE IV

EVALUATION METRICS, INCLUDING THE ACCURACY PER CATEGORY (THE FIRST SIX ROWS), OA, AA, PT, AND NMI, ON M_L WITH A RELATIVELY LARGE GSD IN THE MUUFL DATASET. THE BEST METRICS ARE HIGHLIGHTED IN BOLD. THE OBTAINED CONCLUSIONS ARE FURTHER VERIFIED

Evaluation Metrics	kmeans	SPC	SSC	CLASSIX	BGPC	DGAE	UADNet	3DSSC	SEC-IMVC	ETLSRR	DCCNMF	LATER	EOMSC	MvSCGE	IamCSC
Category 1	28.56	44.15	36.25	43.62	35.09	22.76	31.72	38.88	45.73	51.32	**67.23**	28.45	67.02	41.31	64.49
Category 2	68.24	31.18	68.82	**100.00**	0.00	80.00	94.71	57.06	81.76	0.00	7.65	92.35	0.00	73.53	97.65
Category 3	65.05	82.04	68.93	5.34	**99.51**	42.23	35.92	94.17	49.51	86.89	20.39	41.26	97.57	62.62	8.25
Category 4	70.19	89.81	**94.72**	60.38	87.17	72.08	78.87	87.55	84.53	92.08	17.74	50.57	86.42	63.02	87.55
Category 5	**100.00**	96.05	92.11	0.00	94.74	**100.00**	0.00	98.68	93.42	93.42	6.58	94.74	96.05	50.00	98.68
Category 6	48.57	7.43	0.00	53.14	80.57	21.71	8.00	0.57	4.00	1.71	28.57	78.86	5.71	**82.86**	54.29
AA (%)	63.43	58.44	60.14	43.75	66.18	56.46	41.54	62.82	59.83	54.24	24.69	64.37	58.80	62.22	**68.48**
OA (%)	47.15	52.42	50.19	46.06	53.34	40.41	41.23	52.58	53.07	53.45	43.18	46.50	62.41	54.10	**65.02**
PT (÷100)	65.89	65.29	74.31	68.93	71.32	66.38	64.04	72.46	67.57	70.83	51.55	64.69	71.27	72.41	**75.39**
NMI (÷100)	40.76	43.54	51.35	43.11	49.62	37.52	37.11	51.25	34.28	49.16	6.53	34.28	54.41	46.57	**55.68**

matched with real categories. IamCSC generates better clustering performance and enhances the clustering explainability by distilling an IAM. This explainable fusion strategy is also our contribution considering the specific geological characteristics of HS and LiDAR images. This article aims to demonstrate the challenges faced by these clustering techniques using

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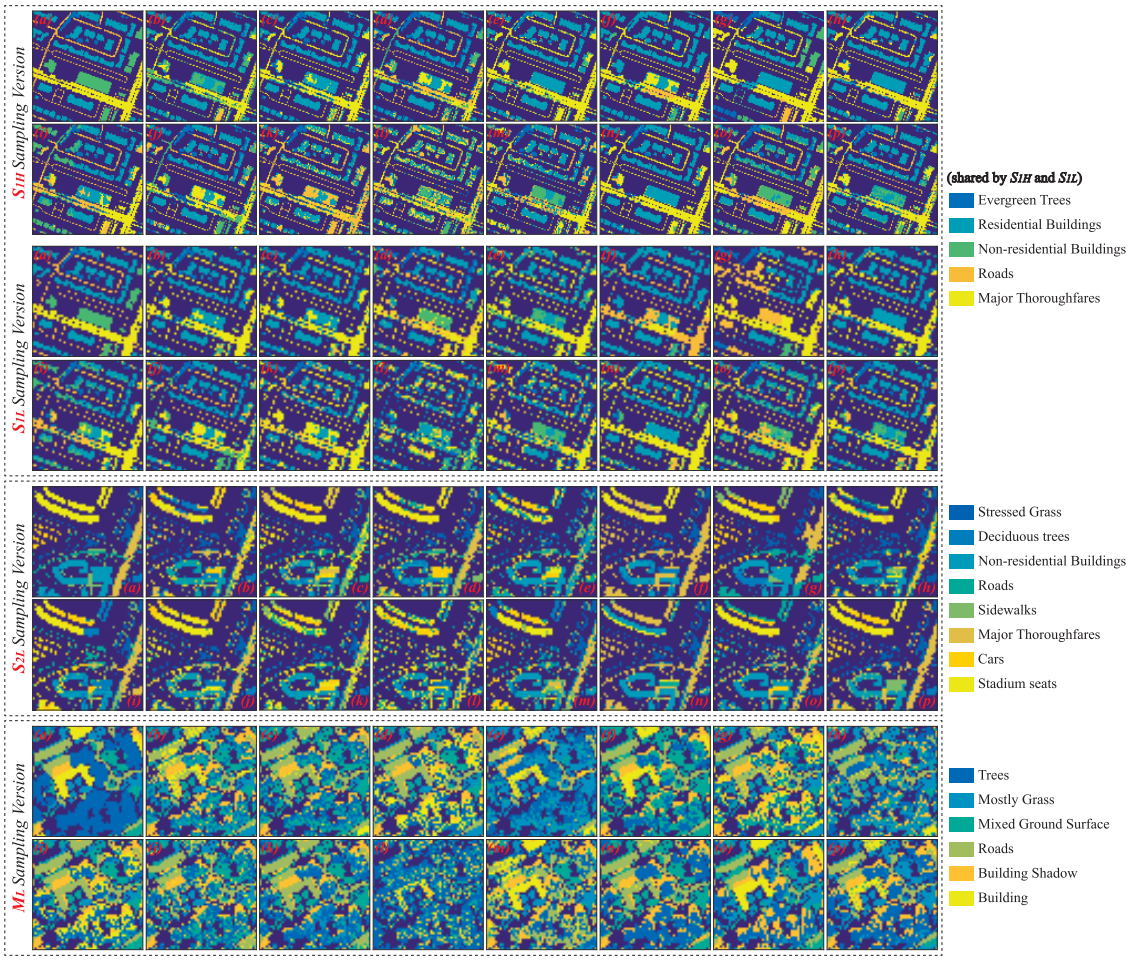


Fig. 4. Clustering maps of (a) ground truth, (b) k-means, (c) SPC, (d) SSC, (e) CLASSIX, (f) BGPC, (g) DGAE, (h) UADNet, (i) 3DSSC, (j) SEC-IMVC, (k) ETLSRR, (l) DCCNMF, (m) LATER, (n) EOMSC, (o) MvSCGE, and (p) IamCSC on S_1H, S_1L, S_2L, and M_L. The dark blue regions are unlabeled.

low-quality RS images. Fig. 3 shows that S_{1L} with a larger GSD tends to lose detailed information compared to S_{1H}. Thus, the S_{1L} results are lower compared to those of S_{1H}. For example, the OA of IamCSC on S_{1L} is 1.21% lower than that on S_{1H}. For S_{1H}, IamCSC outperforms both k-means (the single-view method with the highest OA) and EOMSC (the multiview method with the highest OA) in terms of OA, achieving improvements of 4.65% and 4.12% over these two techniques, respectively. It outperforms other state-of-the-art methods in terms of the OA and PT. However, the best AA and NMI are captured by MvSCGE and EOMSC, respectively, primarily due to their superior performance in the *Evergreen Trees* and *Nonresidential Buildings* regions and the *Major Thoroughfares* region, respectively. Although EOMSC cannot detect any *Roads* samples, the superiority of IamCSC is not fully demonstrated. In contrast, by increasing the GSD, the comprehensive effectiveness of IamCSC can be highlighted across all metrics. Specifically, IamCSC demonstrates superior performance across all metrics compared to both EOMSC and MvSCGE, with improvements of 16.74%, 5.39%, 3.38%, 0.04% and 0.69%, 11.62%, 5.49%, 6.36%, respectively. IamCSC detects both the *Evergreen Trees* and *Residential Buildings* regions simultaneously. These results show that IamCSC exhibits superior advantages when processing RS image pairs with low resolutions. The results on S_{2L} and M_L corroborate the effectiveness of IamCSC in comparison to other approaches. Although IamCSC may not achieve the highest accuracies across all categories, its innovative optimization mechanism and focus on explainability collectively contribute to satisfactory clustering performance.

The clustering maps are provided in Fig. 4. The results show that IamCSC can yield superior qualitative clustering performance compared to other methods, particularly when dealing with low-quality RS images. As shown in Fig. 5, some comparison techniques, e.g., DGAE, UADnet, and EOMCS, fail to detect the *Nonresidential Buildings* regions, as they tend to categorize these areas along with the *Residential Buildings* regions into a single cluster. These maps prove that IamCSC accurately clusters most regions, even when these RS images exhibit relatively high GSDs. Although MvSCGE identifies the largest nonresidential building, it exhibits a tendency to wrongly regard some residential buildings as nonresidential buildings. The reason is that MvSCGE fails to balance the detailed contributions of HS and LiDAR images during the information fusion procedure. Furthermore, IamCSC detects the tree shadow regions in M_L accurately although they are not annotated in the ground truth. In summary, these results conclusively prove the effectiveness of IamCSC, particularly in low spatial resolution RS image pairs with large GSDs.

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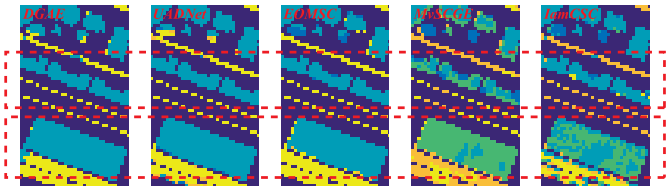


Fig. 5. (Upper box) Enlarged Residential Buildings and (lower box) Non-residential Buildings regions by DGAE, UADnet, EOMCS, MvSCGE, and IamCSC to prove the category identification ability of IamCSC qualitatively.

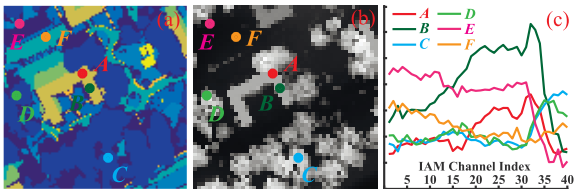


Fig. 6. Visualization of the distilled IAM in IamCSC to verify the clustering explainability, including. (a) Ground truth. (b) LiDAR image. (c) IAM channel curves of six IAM samples. The spatial positions of these IAM samples are marked in both (a) and (b) to reflect their land-cover categories and elevations.

In addition to achieving superior clustering performance, IamCSC offers satisfactory explainability for multimodal clustering by incorporating geological attributes and descriptive features derived from both HS and LiDAR images. Specifically, the distilled IAM reflecting both the HS spectral correlations and LiDAR elevations is obtained by utilizing diverse matrix-wise constraints. Fig. 6 provides a clear visualization of these intuitive capabilities and verifies the feasibility of IamCSC. The IAM channel curves of six IAM samples (the distilled IAM in IamCSC contains d IAM channels, d was set as 40 in these experiments), including A (red, *Building*), B (dark green, *Building*), C (blue, *Trees*), D (light green, *Trees*), E (pink, *Mixed Ground Surface*), and F (orange, *Roads*), are provided. The IAM channel curves of A and B (as well as those of C and D) exhibit remarkable patterns and peak region consistencies considering their shared land-cover category and comparable elevations. On the other hand, while E and F share similar elevations and thus, have consistent peak regions, their curve patterns are inconsistent because they do not belong to the same category. Additionally, the peak regions of these IAM channel curves are indicative of the elevation differences. Notably, larger curve values correspond to higher IAM channel indexes when the sample elevations are big.

D. Parameter Selection

Experiments were conducted on M_L in the Mapping and Urban Unmanned Systems (Florida) Lab dataset to select the optimal parameters in IamCSC, inclu ding the IAM channel number d, the tradeoff parameters $\{\alpha_1, \alpha_2, \beta\}$, the global smoothing index τ and the initial learning rate η.

1) Influence of the IAM Channel Number d: The IAM channel number d is constrained within $[1, b]$ $(d < b)$ since the IAM balances the channel information of the HS image with b HS channels and the LiDAR data with 1 LiDAR channel. For the MUUFL dataset, the optimal d was selected within $\{10, 20, \ldots, 60\}$ since this dataset contains 64 spectral channels $(d < 64)$. Typically, the IAMs with smaller ds can neither clearly preserve HS spectral similarity and diversity nor reflect sample elevations by peak regions. Conversely, IAMs with larger ds tend to overfit the HS images, compromising the assimilation performance. The evaluation metrics with diverse ds are listed in Table V. It shows that the best performance is achieved when $d = 40$, despite a marginal increase in the PT of 0.17% as d rises from 40 to 60. Considering all evaluation metrics comprehensively, $d = 40$ was selected as the default value. When $d > 30$, the accuracies are relatively stable since IamCSC is not sensitive to d. Thus, the IAM channel number d for the Houston 2018 dataset with 48 HS channels was also set as 40 for simplification. Notably, in contrast to band selection, IamCSC aims not at reducing HS channel numbers, but at comprehensively fusing HS and LiDAR images. The reason why d is smaller than b is that IamCSC tends to balance the channel information between the HS image with b HS channels and the LiDAR data with one LiDAR channel. If IamCSC is improved for two different HS images with b_1 and b_2 HS channels $(b_1 < b_2)$ in future works, d should be constrained within $[b_1, b_2]$ correspondingly.

TABLE V

EVALUATION METRICS WITH DIFFERENT dS, INCLUDING OA, AA, PT, AND NMI, ON M_L IN THE MUUFL DATASET

d	10	20	30	40	50	60
AA (%)	64.08	67.77	68.37	**68.48**	68.25	68.21
OA (%)	60.78	63.88	64.10	**65.02**	64.91	64.80
PT (÷100)	72.19	74.20	73.87	75.39	75.29	**75.56**
NMI (÷100)	50.90	54.15	54.22	**55.68**	55.44	55.67

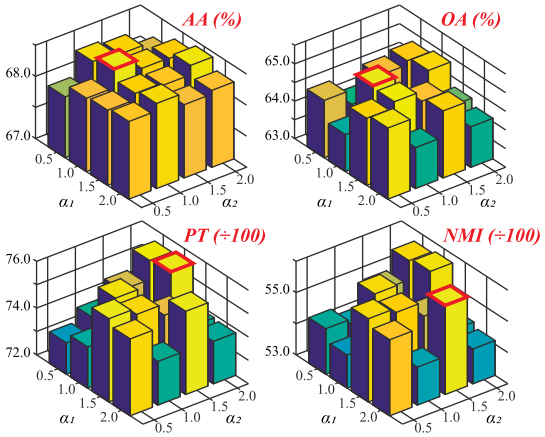


Fig. 7. Evaluation metrics with different α_1s and α_2s, including OA, AA, PT, and NMI, on M_L in the MUUFL dataset. The best results are boxed.

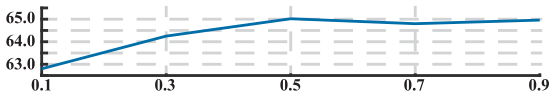


Fig. 8. Rising OA curve for different βs on M_L in the MUUFL dataset.

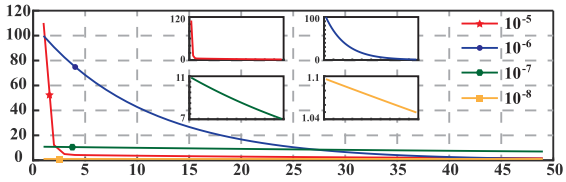


Fig. 9. Convergence curves within the initial 50 epochs with different initial learning rates on M_L in the MUUFL dataset.

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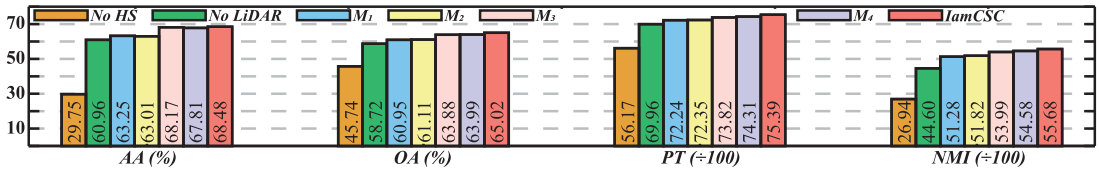


Fig. 10. Evaluation metrics of the alternative scenarios, including OA, AA, PT, and NMI, on M_L in the MUUFL dataset.

TABLE VI
EVALUATION METRICS WITH DIFFERENT τS, INCLUDING OA, AA, PT, AND NMI, ON M_L IN THE MUUFL DATASET

τ	1	3	5	7	9	11	13
AA (%)	68.21	68.48	68.25	**68.50**	68.39	68.37	68.27
OA (%)	64.80	**65.02**	64.91	64.64	64.96	64.10	64.04
PT (÷100)	**75.56**	75.39	75.29	74.85	75.34	73.87	73.82
NMI (÷100)	55.67	**55.68**	55.44	55.25	55.56	54.22	54.11

2) Influence of the Trade-Off Parameters: The tradeoff parameters $\{\alpha_1, \alpha_2, \beta, \lambda\}$ control the constraint contributions in (13). For simplification, λ was set as 0.1 directly. Initially, β was fixed to 0.5 for selecting the optimal α_1 and α_2 within $\{0.5, 1.0, 1.5, 2.0\}$. The evaluation metrics, including AA, OA, PT, and NMI, with diverse α_1 and α_2 couples are provided in Fig. 7. The results prove that the best AA and OA appear when both α_1 and α_2 are equal to 1.0. Although the best PT and NMI can be obtained when α_1 and α_2 are set as $\{1.0, 2.0\}$ and $\{2.0, 1.5\}$, respectively, the corresponding metrics are satisfactory when α_1 and α_2 are set as $\{1.0, 1.0\}$. Thus, the default values of α_1 and α_2 are both 1.0. This conclusion verifies that both constraints, i.e., $\Upsilon_{\text{CG-HS}}(\hat{\mathbf{X}}, \hat{\mathbf{X}}_1)$ and $\Upsilon_{\text{EE-LiDAR}}(\hat{\mathbf{X}}, \hat{\mathbf{X}}_2)$, are equally important for enhancing the clustering capability and explainability. Subsequently, these two parameters were fixed to their defaults to select the optimal β within $\{0.1, 0.3, \ldots, 0.9\}$. The OAs with diverse βs are shown in Fig. 8. Apparently, OA attains its maximum when $\beta = 0.5$ (selected for IamCSC), thereafter exhibiting a tendency toward stability as β increases further. The results also confirm that IamCSC is insensitive to β, providing a broad selection range for β in real-world scenarios.

3) Influence of the Global Smoothing Index τ: The global smoothing index τ was selected within $\{1, 3, \ldots, 13\}$. The evaluation metrics are shown in Table VI. The best clustering performance is yielded when $\tau = 3$ in terms of OA and NMI. Although AA and PT reach their maximum values when $\tau = 7$ and $\tau = 1$, respectively, $\tau = 3$ was selected for IamCSC when considering all evaluation metrics comprehensively.

4) Influence of the Initial Learning Rate η: Generally, the initial learning rate η primarily influences the convergence speed rather than the clustering performance. In this article, η was selected within $\{10^{-5}, 10^{-6}, 10^{-7}, 10^{-8}\}$ to analyze its impact on the entire convergence procedure. The convergence curves over the initial fifty epochs are provided in Fig. 9. These curves are individually plotted to illustrate their convergence patterns clearly. When compared to the other two curves, the curves corresponding to $\eta = 10^{-5}$ and $\eta = 10^{-6}$ rapidly attain stability. Consequently, the default value of η was set as 10^{-5}.

E. Ablation Study

1) Contributions of Both RS Modalities: The contributions of both HS and LiDAR images are comprehensively analyzed by presenting two alternative scenarios including **No HS** and **No LiDAR**. In these scenarios, the HS and LiDAR images, respectively, along with their corresponding constraints, are excluded. The evaluation metrics are listed in Fig. 10. The results show that the clustering performance deteriorates when removing either RS modality, thereby confirming that both HS and LiDAR images contribute to achieving satisfactory land-cover identification results. When omitting the HS image, these metrics decrease by 38.73%, 19.28%, 19.22%, and 28.74%, respectively. Similarly, when removing the LiDAR image, the metrics drop by 7.52%, 6.3%, 5.43%, and 11.08%, respectively. These scenarios validate the necessity of considering multisource RS images in real-world tasks.

2) Contributions of the Matrix-Wise Constraints: In (13), some matrix-wise constraints are used to enhance its clustering capability and explainability. To verify their contributions, several alternative scenarios are designed, including M_1 (when $\alpha_1 = 0$), M_2 (when $\alpha_2 = 0$), M_3 (when $\beta = 0$), and M_4 (when $\lambda = 0$). Their other settings are consistent with those in IamCSC. The evaluation metrics are provided in Fig. 10. Initially, M_1 and M_2 exhibit similar performance in terms of these metrics, further verifying that both the CG-HS and EE LiDAR constraints are equally important for increasing the capability of IamCSC. This conclusion is consistent with the obtained one in Section IV-D2. Compared to M_3 and M_4, IamCSC increases the OA by 1.14% and 1.03%, respectively. Although the enhancement is relatively modest, it remains useful in demonstrating the effectiveness of IamCSC.

V. CONCLUSION

In this article, a multiview SC method called IamCSC has been proposed for land-cover identification and HS-LiDAR fusion when annotated samples are unavailable. This method aims to exploit intuitive crossmodal information and offer an interpretable fusion mechanism. Compared to existing multiview clustering approaches, IamCSC contributes to distilling an IAM with distinguishing geological characteristics of both HS and LiDAR images simultaneously. The distilled IAM intuitively as follows: 1) preserves spectral similarity and diversity of neighboring samples and 2) reflects sample elevations by peak regions. It is regarded as the shared information source and is capable of being mapped back to the original RS modalities. Furthermore, it connects and balances both RS modalities to capture an assimilation cluster structure from modality-specific subspace representations. Diverse matrix-wise constraints have been established for optimizing the optimal modality-shared representation. Experiments prove that IamCSC outperforms several state-of-the-art single-view and multiview clustering techniques qualitatively and quantitatively. Hampered by high orbit altitudes and limited focal lengths, real-world RS images always have low resolutions

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and blurred details, hindering the improvement of clustering performance. IamCSC does not exhibit over-reliance on spatial neighborhood contributions, making it capable of achieving satisfactory performance in low spatial resolution RS image pairs. Further experiments will focus on simulating this explainable multiview clustering mechanism for enhancing the clustering performance.

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